

Jalpaiguri Govt. Engg. College
(A Govt. Autonomous College)
COE/B.Tech/CSE/PCC-CS504 /2025-26
2025

FM: 70

OPERATING SYSTEM

Time Allotted: 3 hours

The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

Group – A

[Objective Type Questions]

Answer all questions

5 X 2 = 10

1. "Switching is pure overhead to the system". Explain
2. What is page fault?
3. What is the role of program counter?
4. What is Roll in and Roll out in swapping?
5. Write the difference between job scheduler and CPU scheduler.

Group – B

[Long Answer Type Questions]

Answer any four of the following

4 X 15 = 60

6. a. What is Process Control Block? What is the information stored here?
b. Consider a logical address space of 8 pages of 1024 bytes each, mapped onto a physical memory of 32 frames. How many bits are there in the logical address? How many bits are there in the physical address?
c. What is FCFS scheduling? Suppose that the processes arrive in the order: P_1, P_2, P_3 with CPU Burst time of 18, 6, 3. With a Gantt chart find waiting time of every process and also find average waiting time.
5 + 5 + (1+2+2)
7. a. What is Thread? Write the difference between Process and Thread?
b. Why is the page table size chosen to be a power of 2? Can you the size be a power of 3?
c. Discuss the difference between Paging and Segmentation in details.
(1+4) + 5+5
8. a. Explain how a scheduling algorithm is called preemptive or non-preemptive with example.
b. What will happen in Round Robin scheduling if the time quantum is zero? Is the system will be in deadlock or not? Justify your answer.
c. Draw the diagram of paging hardware and discuss how it works.
5+5+5
9. a. Describe storage hierarchy with a suitable diagram
b. What are the advantages and disadvantages of preemptive scheduling over non-preemptive scheduling?
c. Discuss different address binding scheme in details? Out of this which is mostly used and why?
5 + 5+ 5
10. a. Discuss about different scheduling criteria.
b. Write short notes on Virtual Memory.
c. What is Belady's Anomaly? Give one example taking reference pages as 1,2,3,4,1,2,5,1,2,3,4,5, where the Belady's Anomaly occurs.
5+ 5+5

PTO

11. a. What are the necessary and sufficient conditions for deadlock? Describe them.
b. What is semaphore? Why signal () and wait () are called atomic operations?
c. Discuss Least Recently Used (LRU) Algorithm with Example.

5 + 5 + 5

12. Write short notes on-
- a. Different Process State
 - b. Dynamic Storage-Allocation Problem
 - c. Logical vs. Physical address space.
 - d. Demand Paging
 - e. Internal vs. External Fragmentation.

5x3=15

END

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
[A GOVERNMENT AUTONOMOUS COLLEGE]
JGEC/B.TECH./CSE /PCC-CS-502 2025-2026
2025

Discrete Mathematics

Full Marks: 70

Times: 3 Hours

The figures in the margin indicate full marks.
Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A		
[OBJECTIVE TYPE QUESTIONS]		
Answer all questions		5x2=10
1.	Find the numeric sequence of the generating function $(z^6+1)/(2z+1)$, where $z < 1/2$.	2
2.	State Pigeonhole Principle, with example?.	2
3.	Draw the Hasse diagram of PO set $(S = \{1, 2, 3, 8, 12, 18, 36\}, /)$. Where / stand divisibility.	2
4.	Using principle of mathematical induction prove that $3^n + 7^n - 2$ is divisible by 8, Where $n \in \mathbb{Z}$.	2
5.	Show that the mapping $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3x + 5$, $x \in \mathbb{R}$ is bijective mapping.	2
GROUP-B		
[LONG ANSWER TYPE QUESTIONS]		
Answer any four questions		4x15=60
6.	i) If $f, g: \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = ax + b$, $g(x) = 1 - x + x^2$ and $(g \circ f)(x) = 9x^2 - 9x + 3$, find the values of a, b .	5
	ii). Show that a group $(G, 0)$ is abelian if and only if $(a \circ b)^2 = a^2 \circ b^2$, $a, b \in G$.	5
	iii) Prove that every Field is an integral domain.	5
7. i)	(i) Five dice are thrown simultaneously. In how many ways a total of 16 can be obtained?	6
	ii) State Euclid's Algorithm. Using this Algorithm find the G C D $(-1479, 272)$.	5
	iii) Find the missing digit in the number $8630480*874$ if it divisible by a) 11, b) 7.	4
8. i)	Find the remainder when 2^{230} is divided by 341.	4
	ii) How many ways the letters of the word "Allahabad" be arranged in each of which the vowels (a) never occur together (b) occupy odd positions.	2 +
	iii) Solve the recurrence relation $a_n - 5a_{n-1} + 6a_{n-2} = 2^n + n$ with $a_0 = 1, a_1 = 1, n \geq 2$ using generating function.	2 7
9. i)	Using principle of mathematical induction prove that $(2 + \sqrt{3})^n + (2 - \sqrt{3})^n$ is an even integer.	5
	ii) Show that every tree with two or more vertices is 2-chromatic.	5
	(iii) Show that $((p \vee q) \wedge (\sim p \wedge (\sim q \vee \sim r))) \vee (\sim p \wedge \sim q) \vee (\sim p \wedge \sim r)$ is a tautology.	5
10. i)	Solve $u_{n+2} - 6u_{n+1} + 9u_n = n3^n$.	5
	ii) Find the total number of distinct divisors of 444720375.	4

	iii) Find the shortest path from V to W using Dijkstra's Algorithm.	6
11	i) Each user of a computer system has a password which is six to eight characters long, where each character is an upper or lower case English letter or a special character from the set {#,*,\$}. Each password must contain at least one letter and one digit and one special character. How many possible passwords are there?	6
	ii) Find the total number of integers between 1 and 2000 that are not divisible by the integers 2, 3, 7.	4
	iii) Consider the statements p: "I will study discrete mathematics", q: "I will go to movie", r: "I am in a good mood". Write the following statements in terms of p, q, r and logical connectives. Construct the truth table for each of them. "If I am not in good mood then I will go to a movie, else I will study discrete mathematics."	5
12	(i) Find the minimum number of students in M.Tech. first semester to be sure at least six will receive the same grade, if there be five possible grades A, B, C, D and F.	5
	ii) Find the Remainder when 3^{287} is divided by 23.	4
	iii) Write CNF and DNF of $p \rightarrow (p \rightarrow q) \wedge \sim(\sim q \vee \sim p)$.	3 + 3

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
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JGEC/B.TECH/CSE/PCC-CS503/2025-26
2025
DATABASE MANAGEMENT SYSTEMS

Full Marks: 70

Times: 3 Hours

The figures in the margin indicate full marks.
Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

5x2=10

Answer *all* questions

1. Define derived attribute with example.
2. What is weak entity set? Give example.
3. What is meta data? Give example.
4. Define spurious tuple with an example.
5. Define foreign key with example.

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

4x15=60

Answer any *four* questions

6. i) What is referential integrity? State the SQL / MySQL query to define the referential integrity in a database. 3

ii) Consider the following three tables.

Sailor (SID, Sname, Rating, Age)

Reserve (SID, BID, Date)

Boat (BID, Bname, Color)

Express the following queries in Relational Algebra and draw the query tree for the respective queries.

a) Find Sailors' name with age > 50 who've reserved a red or a green boat.

b) Find Sailors' name who've reserved atleast one red and one green boat.

c) Find the names of Sailors who've reserved all boats.

12

- i) Explain the distinctions among the terms primary key, candidate key and super key.. 5
- ii) Define composite attribute and multi-valued attribute. Give two real-life examples for each of them. Which type of attribute should be avoided to design a 1NF relation? 5
- iii) Find the canonical cover of the functional dependency set $F = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$. 5

i) Find the candidate keys of the following relations: 5

a) $R1 = (B, O, S, Q, I, D); F = \{S \rightarrow D, I \rightarrow B, IS \rightarrow Q, B \rightarrow O\}$

b) $R2 = (S, D, M, I); F = \{SI \rightarrow D, SD \rightarrow M\}$

c) $R3 = (C, T, H, R, S, G); F = \{C \rightarrow T, HR \rightarrow C, HT \rightarrow R, CS \rightarrow G, HS \rightarrow R\}$

- ii) Determine the functional dependencies of the following table and also identify the highest normal form of the given relation.

STUDENT	COURSE	INSTRUCTOR
Nasir	dbms	Mark
Smith	dbms	Navathe
Smith	os	Ani
Smith	theory	Sami
Robin	dbms	Mark
Robin	os	Akki
Bonnie	dbms	Tressy
Zaid	dbms	Navathe

- iii) Consider the following relation and functional dependency set.

$R = (C, T, H, R, S, G); F = \{C \rightarrow T, HR \rightarrow C, HT \rightarrow R, CS \rightarrow G, HS \rightarrow R\}$

Decompose the relation into BCNF relations.

9. i) Compute the closure of attribute A from the given set F of functional dependencies $F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$ for relation schema $R = (A, B, C, D, E)$. Is 'A' a candidate key for R? Which attribute of R has the same closure value as 'A' closure? List all the candidate keys of R. 6
- ii) Define different anomalies in a relational database. "Anomalies are related to each other." - Explain. 5
- iii) Differentiate between a traditional file processing system and a database system. 4
10. i) Describe the ACID properties of transactions with suitable examples. 5
- ii) Draw the state diagram of a transaction execution and explain. 5
- iii) Consider the schedule S1 given below. Is this schedule S1 serializable? Draw the precedence graph and explain it. 5

T1	T2	T3
	R(Z) R(Y) W(Y)	
		R(Y) R(Z)
R(X) W(X)		
		W(Y) W(Z)
	R(X)	
R(Y) W(Y)		
	W(X)	

11. i) What are the differences between B tree and B + tree index structures? 2
- ii) Suppose a table contains PartNo as a key field and the records in that table with the following PartNo values 8, 5, 1, 7, 3, 12, 9, 6. Suppose that the search field values are inserted in the given order in a B + tree of order $p=3$ and $p_{leaf}=2$; Show how the tree will expand and what the final tree will look like. 5
- iii) Now delete 5, 12 from the above B+ tree and show the updated tree. 3
- iv) Explain different types of outer join with example. 5
12. Write short notes on the following 5
- i) Lossless join decomposition and dependency preservation 5
- ii) Set operations in relational DBMS 5
- iii) Recoverable schedule and cascade less schedule
-

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
[A GOVERNMENT AUTONOMOUS COLLEGE]
JGEC/B.TECH/ CSE / PCC-CS501 / 2025-26
2025

SOFTWARE ENGINEERING

Times: 3 Hours

Full Marks: 70

*The figures in the margin indicate full marks.
 Candidates are instructed to write the answers in their own words as far as practicable.*

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

5x2=10

Answer **all** questions

1. Define Data Dictionary and its importance. 2
2. What do you mean by size of a software product? How can project size be estimated? 2
3. What is a context diagram? 2
4. What do you mean by software metrics? Give two examples. 2
5. "System testing can be considered a pure black-box testing". Justify the answer. 2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

4x15 = 60

Answer any **five/four** questions

6. i) What is a prototype? Draw a schematic diagram and explain the prototyping model of software development. Differentiate between prototyping model and evolutionary model. 1+4+3
- ii) According to Boehm's, explain different types of software products with examples? Differentiate basic COCOMO with complete COCOMO. 3+2
- iii) "Spiral lifecycle model can be considered to be a Meta model"-- justify 2
7. i) If a software product for business application costs Rs. 1, 00,000 to buy and that its size is 60 KLOC assuming that in house developers cost Rs, 10,000 per month. Which one is more cost-effective to buy or build the product? 6
- ii) Using Putnam's method, show how effort required to develop a project depends on project schedule. 3
- iii) Write the different types of risks that software might suffer from. Explain three essential activities for risk management. 3+3
8. i) Mention the importance of phase containment of errors. Write the main differences between 'structured chart and flow chart' in the context of software design. 2+3
- ii) Define the significance of cohesion and coupling for a good software design. Explain different types of cohesion. 3+5
- iii) Write down the roles of SRS documents in the life cycle model of software development. 2
9. i) Define software reliability and software quality. Explain the different levels of SEI-CMM with their key process areas. 3+5
- ii) Explain using one simple sentence each what you understand by the following reliability measures: 4
 (a) A POFOD of 0.002 (b) A ROCOF of 0.005 (c) MTBF of 200 units (d) Availability of 0.998
- iii) What do you understand about Quality Assurance and Total Quality Management? 3

60

10. i) Write down the importance of PERT chart and GNATT chart. Define critical path. 4+1
- ii) Write down the different types of Project estimation techniques. Estimate the different project parameters using Halstead Software Science. 3+5
- iii) Define software maintenance. 2
11. i) Distinguish between software verification and software validation. Differentiate between structural complexity and computational complexity of a program. 3+3
- ii) Draw the control flow graph for the function find largest, which will find the largest number from a set of input numbers. From the control flow graph determine its cyclometric complexity. 3+2
- iii) Write how white box testing is different from black box testing. 4
12. Write Short Notes on any three
- i) UML ii) Data flow based testing iii) Decision Tree and Decision Table iv) Spiral Model
- v) Integration Testing

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
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JGEC/B.TECH/ CSE/PEC-CS501D/ 2025-26
2025
Operation Research

Full Marks: 70

Times: 3 Hours

The figures in the margin indicate full marks.
Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer *all* questions

1. What do you mean by degeneracy in Transportation problem?
2. What is slack variable?
3. What do you mean by burst event in a network (explain with figure)?
4. What do you mean by unbalance assignment problem? How will you solve the problem?
5. Which queuing model is called birth and death model?

5x2=10

2
2
2
2
2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any *five/four* questions

4x15 = 60

7

- ~~6.~~ i) Use the graphical method to solve the following Linear programming problem

Minimize $Z = 20x_1 + 10x_2$
 Subject to constraints

$$\begin{aligned} x_1 + 2x_2 &\leq 40 \\ 3x_1 + x_2 &\geq 30 \\ 4x_1 + 3x_2 &\geq 60 \\ x_1, x_2 &\geq 0 \end{aligned}$$

- ii) Use simplex method to solve the following LPP
 Maximize $Z = 15x_1 + 6x_2 + 9x_3 + 2x_4$
 Subject to constraints

8

$$\begin{aligned} 2x_1 + x_2 + 5x_3 + 6x_4 &\leq 20 \\ 3x_1 + x_2 + 3x_3 + 25x_4 &\leq 24 \\ 7x_1 + x_4 &\leq 70 \\ x_1, x_2, x_3, x_4 &\geq 0 \end{aligned}$$

7. i) Solve the given transportation table using a) North-West corner) Least cost and c) VAM and calculate total transportation cost in each case

(4+4+4)

	1	2	3	Availability
A	5	2	7	9
B	9	15	10	6
C	8	18	5	5
Requirement	5	5	10	

- ii) Explain allocations in independent position in transportation problem

3

8. i) By the notion of dominance, reduce the game and Solve the game graphically

9

Player A	Player B	
	B1	B2
	A1	1 -3
	A2	3 5
	A3	-1 6
	A4	4 1
	A5	2 2
	A6	-5 0

- ii) The standard weight of a special purpose brick is 5 kg and it contains two ingredients B1 and B2. B1 costs Rs 5 per kg and B2 costs Rs 8 per kg. Strength considerations dictate that the brick contains not more than 4 kg of B1 and a minimum of 2 kg of B2 since the demand for the product is likely to be related to the price of the brick. Formulate the above problem as a L. P. Model

6

9. i) Tasks A, B,.....H,I constitute a project. The notation $X < Y$ means that X task must be completed before Y has been started. With the notation $A < D, A < E, B < F, D < F, C < G, C < H, F < I, G < I$
Completion time of each task is given below

(3+6+2)

Task	A	B	C	D	E	F	G	H	I
Time(days)	8	10	8	10	16	17	18	14	9

Draw the network diagram and find the minimum time of completion of the project. Also calculate total float of each non critical activity

- ii) Define three types of time estimate in PERT technique. How to calculate expected time from these time estimates:

4

10. i) Using the following cost matrix, determine (a) optimal job assignment (b) the cost of assignments

7

Jobs	Machines				
	A	B	C	D	E
1	13	8	16	18	19
2	9	15	24	9	12
3	12	9	4	4	4
4	6	12	10	8	13
5	15	17	18	12	20

- ii) Following Payoff Matrix is given

6

Player A	Player B				
		B1	B2	B3	B4
	A1	1	7	3	4
	A2	5	6	4	5
	A3	7	2	0	3

Determine the best strategies and also the value of the game. Is this game (i) fair (ii) strictly determinable?

2

- iii) What is the purpose of MODI method?

11. i) Use Big 'M' method to solve the following problem

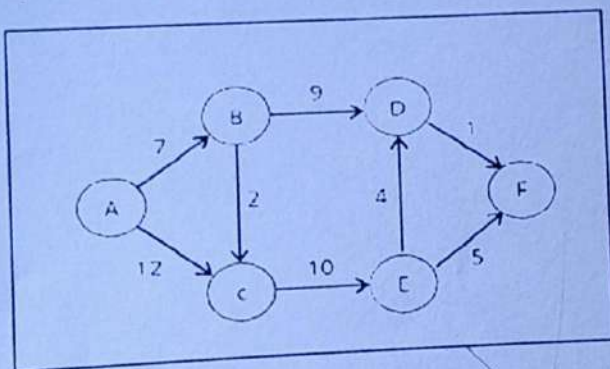
$$\text{Maximize } Z = 3x_1 + 2x_2$$

Subject to constraints

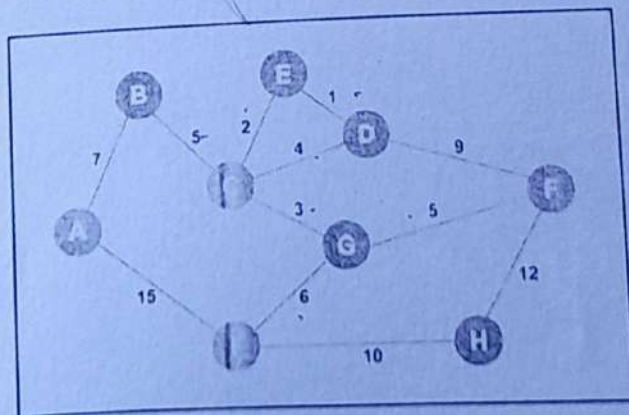
$$\begin{aligned} 2x_1 + x_2 &\leq 2 \\ 3x_1 + 4x_2 &\geq 12 \\ x_1, x_2 &\geq 0 \end{aligned}$$

ii) Explain Kendall's notation for representing queueing model

12. i) Using Dijkstra's algorithm find the length of the shortest path from source A to each vertex



ii) Find the minimum spanning tree calculate its duration using Kruskal's algorithm. Calculate the minimal cost



iii) What do you mean by traffic intensity in a queue(Give the expression)

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
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JGEC/B.TECH/CSE/PEC-CS501B/2025-26
2025
SOFT COMPUTING

Times: 3 Hours

Full Marks: 70

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Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

5x2=10

Answer **all** questions

- What is the output of a fuzzy inference system before de-fuzzification?
- What is singleton?
- What is Artificial Neural Network?
- In what ways is bipolar representation better than binary representation?
- Define bias and threshold.

2
2
2
2
2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

4x15=60

Answer any **four** questions

6. a. Let $U = \text{Fruits} = \{A, B, C, D, E, F\}$ be a universe on which two fuzzy sets, one of Ripped fruits and the other one of

Fragrant Fruits are defined as shown below:

$$P = \text{Ripped Fruits} = 0.3/A + 0.9/B + 1.0/C + 0.7/D + 0.5/E + 0.4/F + 0.6/G$$

$$Q = \text{Fragrant Fruits} = 1.0/A + 1.0/B + 0.5/C + 0.2/D + 0.2/E + 0.1/F + 0.4/G$$

Compute the Fuzzy sets $P \cup Q, P \cap Q, P', Q', P - Q, P \oplus Q$.

5

- b. Write a short note on fuzzy if-then-else rule.

5

7. a. Define Fuzzy Inference System.
b. Differentiate between crisp and fuzzy set.
c. What do you mean by Generalized Modus ponens?

5

5

5

8. Let $U = \text{Flowers} = \{\text{Jasmine, Rose, Daffodil, Sunflower, Hibiscus, Chrysanthemum}\}$ be a universe on which two fuzzy sets, one of Beautiful Flowers and the other one of Fragrant flowers are defined as shown below:

$$P = \text{Beautiful Flowers} = 0.3/\text{Jasmine} + 0.9/\text{Rose} + 1.0/\text{Lotus} + 0.7/\text{Daffodil} + 0.5/\text{Sunflower} + 0.4/\text{Hibiscus} + 0.6/\text{Chrysanthemum}$$

$$Q = \text{Fragrant Flowers} = 1.0/\text{Jasmine} + 1.0/\text{Rose} + 0.5/\text{Lotus} + 0.2/\text{Daffodil} + 0.2/\text{Sunflower} + 0.1/\text{Hibiscus} + 0.4/\text{Chrysanthemum}$$

Compute the Fuzzy sets $P \cup Q, P \cap Q, P', Q', P - Q, P \oplus Q$.

- Show how multilayer perceptron overcome the classification problem the patterns posed by the XOR problem.

5

$P \cap Q = 0.7, 0.1, 0.1, 0.2, 0.2, 0.1, 0.4$

PTO

8. a. Discuss about different types of '**Learning Techniques**' in machine learning with example. 5
- b. What is activation function? Discuss about, Sigmoid function, Hyperbolic Tangent Function and the Rectifier linear unit with diagram. 5
- c. Define & differentiate "HARD vs SOFT COMPUTING". 5
9. a. Develop the training algorithm to classify the logical '**AND**' function using perceptron learning rule. 10
- b. Show a proper truth table of the training. 5
- c. Define perceptron learning rule. Also inform that in which way it is better than Hebb's learning rule? 5
10. a. Explain in detail the architecture of Mc Culloch - Pitts neuron model and also realize **OR** gate using the above neuron model. 8
- b. Show how multilayer perceptron overcome the classification problem the patterns posed by the **XOR** problem. 7
12. a. With a neat flowchart, explain the simple form of Genetic Algorithm. 5
- b. Write the difference between **Fuzziness vs. Probability** and **Fuzzy Set vs. Crisp Set**. 5
- c. Discuss about different types of **Transformation** applied to fuzzy membership functions. 5

****END****

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
 [A GOVERNMENT AUTONOMOUS COLLEGE]
JGEC/B.TECH/ CSE / PEC-CS502C / 2025-26
2025

ARTIFICIAL INTELLIGENCE

Full Marks: 70

Times: 3 Hours

*The figures in the margin indicate full marks.
 Candidates are instructed to write the answers in their own words as far as practicable.*

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer **all** questions

1. What is a rational agent? Write three factors that make an agent a rational agent.
2. Write down the properties of a searching algorithm.
3. Write down the principle of MNIMAX Algorithm.
4. Explain the different components of AI agent.
5. Which kind of problems can use uninform search? Why it can't be used in all AI problems?

5x2=10

2
2
2
2
2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any **five/ four** questions

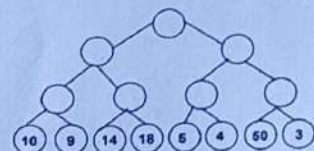
6. (a) Explain bidirectional search technique with proper example.
- (b) Define heuristic function and admissible heuristic with example.
- (c) Define inform search and uninform search. Write down A* algorithm. Is it optimal under all conditions?

12x5 / 4x15 = 60

5
3

3+3+1

7. (a) Solve the following crypt arithmetic problem: EAT + THAT = APPLE
- (b) Compute MINIMAX(root) of the given Game tree:
- (c) What will be the time complexity of MINIMAX algorithm.
- (d) Write down the features of adversarial search and its application.

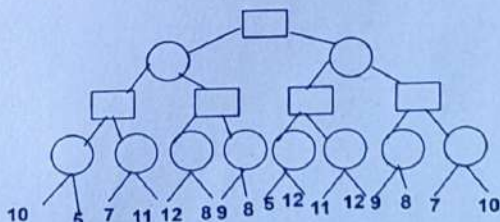


6
5
1
3

8. (a) Explain three properties of AI environment with examples.
- (b) Explain the functions of different components of the learning agent with a proper diagram.
- (c) Write down the differences between goal-based agent and utility-based agent.

6
6
3

9. (a) Write down the principles of local search algorithm. Explain a local search algorithm and its limitations.
- (b) For the given game tree find the value of Alpha of root node using Alpha-Beta pruning algorithm



2+5
8

10. (a) List the requirements for knowledge representation in AI.
- (b) You are given a 4-litre jug and 3-litre jug. Neither has a measuring on it. You have to measure exactly 2 litres of water in the 4-litre jug. Define the production rules and draw the state-space diagram for solving the problem.
- (b) Describe supervised, unsupervised and reinforcement learning.

3
6
6

11. (a) Lists the applications of Natural Language Processing(NLP). Explain the different phases of NLP.
- (b) What do you mean by reasoning? Explain different types of reasoning with examples.
- (c) Explain the basic rules in First Order Logic.

2+6
1+4
2

12. (a) Distinguish between data driven search and goal driven search strategies.
- (b) Consider the following Assumptions: (i) Fido is a dog (ii) All dogs are animals (iii) All animals will die

3
4

Proof that Fido will die using forward chaining

- (c) Represent the following facts in predicate logic: (i) every child likes every candy (ii) every student walks or talks (iii) there is a country that borders both India and Nepal (iv) anyone who loves some candy is not a nutrition fanatic. 4
- (d) Mention two methods of reasoning in uncertain domain. Define a Fuzzy set with example. 2+2
-